Practical 08

Aim: A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Least recently used page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

Code:
```
#include<stdio.h>
int main(){
int p[12]={1,2,3,4,1,2,5,1,2,3,4,5},f[3],i,j,k=0,flag,fault=0;
for(i=0;i<3;i++) f[i]=-1;
printf("Page\tStatus\tFrames\n");
for(i=0;i<12;i++){
flag=0;
for(j=0;j<3;j++)
if(f[j]==p[i]){flag=1;break;}
if(flag) printf("%d\tHit\t",p[i]);
else{
printf("%d\tFault\t",p[i]);
f[k]=p[i];
k=(k+1)%3;
fault++;
}
for(j=0;j<3;j++) printf("%d ",f[j]);
printf("\n");
}
printf("\nTotal Page Faults=%d",fault);
```

return 0;

}

OUTPUT:

```
Page    Status  Frames
1   Fault   1 -1 -1
2   Fault   1 2 -1
3   Fault   1 2 3
4   Fault   4 2 3
1   Fault   4 1 3
2   Fault   4 1 2
5   Fault   5 1 2
1   Hit 5 1 2
2   Hit 5 1 2
3   Fault   5 3 2
4   Fault   5 3 4
5   Hit 5 3 4

Total Page Faults=9
```